Can We Anticipate a High-Risk AQHI Hour?

A Comparative Study of Classification Models on Hourly Monitoring Data

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1. Background and Motivation

Air pollution can change quickly, while Hong Kong's Air Quality Health Index (AQHI) is reported hourly and an AQHI of 7 or above signals high health risk for sensitive groups. The Environmental Protection Department's scheduled forecasts provide useful guidance, but may not fully capture rapid between-forecast changes. This project will test whether a one-hour-ahead warning based on recent station readings and time-available context can provide a useful additional signal, while making the cost of missed warnings versus false alarms explicit.

2. Research Questions

- **RQ1.** How well can lagged AQHI observations, temporal patterns and safely available weather information predict whether a station's AQHI will reach 7 or above in the next hour?
- **RQ2.** How do Logistic Regression, Random Forest and XGBoost compare with majority-class and persistence baselines in detecting rare high-risk hours, and what false-alarm trade-offs result?
- **RQ3.** Which recent AQHI, time and station-level features contribute most to predictions, and are findings stable across roadside and general monitoring stations?

3. Data

- **Sources and access.** Hourly station AQHI from the Environmental Protection Department (EPD) historical archive [1], and daily temperature, relative humidity, rainfall and wet-bulb temperature from the Hong Kong Observatory (HKO) [2]. Public CSV data.
- **Unit and size.** One row represents one station-hour. The processed 2021-2024 dataset contains approximately 617,800 usable rows and 27 candidate predictors from 18 stations (15 general and 3 roadside), spanning 1 January 2021 to 31 December 2024.
- **Response.** Binary next-hour high-risk status: 1 when the same station's AQHI at hour t+1 is >=7, otherwise 0. The positive class is approximately 1.6% of observations.
- **Predictors.** AQHI lags (1, 2, 3, 6, 12 and 24 hours), lagged rolling averages/variability and recent changes; cyclical hour, weekday and month features; station type; and HKO daily weather variables. The primary forecast will include only weather data demonstrably available before the forecast time; daily observations will be used only if publication timing can be confirmed.
- **Known issues and safeguards.** Rare high-risk events, missing station-hours and repeated measurements by station require care. Daily weather is temporally and spatially coarser than AQHI; same-day finalized aggregates may include information unavailable at the forecast hour, so they will be excluded from the primary real-time-feasible model and examined only as a retrospective sensitivity analysis. Data after the March 2025 AQHI methodology change are excluded for comparability. Missing predictors will be handled using training-fold-only rules; future AQHI values will never enter features.

4. Proposed Methods

- **Baseline.** Compare a majority-class predictor and a persistence rule that flags an alert when the current station AQHI is already >=7. These establish whether learned models add value beyond prevalence and immediate persistence.
- **Interpretable model.** Class-weighted Logistic Regression with L2 regularization; use standardized predictors and inspect coefficient direction and magnitude.

- **Flexible models.** Class-weighted Random Forest and XGBoost with positive-class weighting to address imbalance. Tune model settings using training data only.
- **Feature construction.** Build all lags and rolling statistics within station after sorting by time; use only observations available at the forecast origin. Compare feature groups and roadside/general station results.

5. Validation and Evaluation

- **Split and leakage control.** Reserve the latest 20% of unique timestamps as a final chronological test set; keep every station row at the same timestamp in the same partition. Use five expanding-window time-series folds within the earlier training period for model selection. Fit preprocessing and tune alert thresholds inside training folds only; evaluate the holdout once after choices are frozen.
- **Metrics and decision trade-off.** Report precision, recall, F1, PR-AUC and confusion matrices, alongside the class-prevalence baseline. Because missing a high-risk hour is more serious than a false alarm, compare a pre-specified false-negative cost five times the false-positive cost (FN:FP = 5:1), and show threshold sensitivity rather than relying only on 0.5.
- **Interpretation and robustness.** Use logistic coefficients and permutation importance for feature interpretation. Check stability by station type, compare weather-feature timing rules, and report uncertainty using time-block resampling where feasible. No causal claims will be made.

6. Expected Contribution

We expect to establish whether one-hour-ahead classification adds practical warning value over simple baselines, and to quantify how much recall can be gained at different false-alarm rates under severe class imbalance. The analysis can inform how public-health users assess supplementary alerts and which data streams merit closer monitoring; it is not intended to replace official EPD forecasts. Limitations include the rarity of high-risk hours, daily weather resolution and spatial mismatch, possible station-level heterogeneity, and the exclusion of post-2024 observations following the AQHI methodology change. Results will be predictive and observational, not causal.

7. Timeline

Week	Dates	Milestone
1	5-8 Nov 2026	Audit sources and missingness; finalize leakage-safe features, target and exploratory summaries.
2	9-15 Nov 2026	Build majority/persistence baselines and fit Logistic Regression, Random Forest and XGBoost with time-series validation.
3	16-22 Nov 2026	Compare models and alert thresholds; complete interpretation, station-type checks and robustness analyses; freeze results.
4	23-28 Nov 2026	Prepare slides, 8-minute recording and code appendix; quality-check and submit

References

[1] Environmental Protection Department, Hong Kong. Historical Air Quality Health Index (AQHI) data by monitoring station, 2021-2024. https://www.aqhi.gov.hk/epd/ddata/html/history/2021/202101_Eng.csv (archive pattern; accessed 4 November 2026).

[2] Hong Kong Observatory. Daily climate data (mean temperature, relative humidity, rainfall and wet-bulb temperature), HKO Climate Information Services. https://data.weather.gov.hk/weatherAPI/cis/csvfile/HKO/ALL/ (accessed 4 November 2026).

[3] Environmental Protection Department, Hong Kong. About the Air Quality Health Index. https://www.aqhi.gov.hk/en/what-is-aqhi/about-aqhi.html (accessed 4 November 2026).